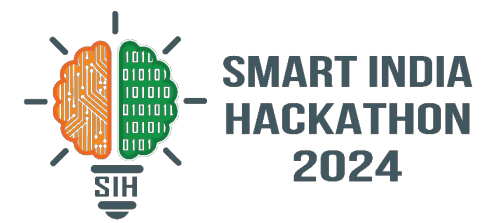
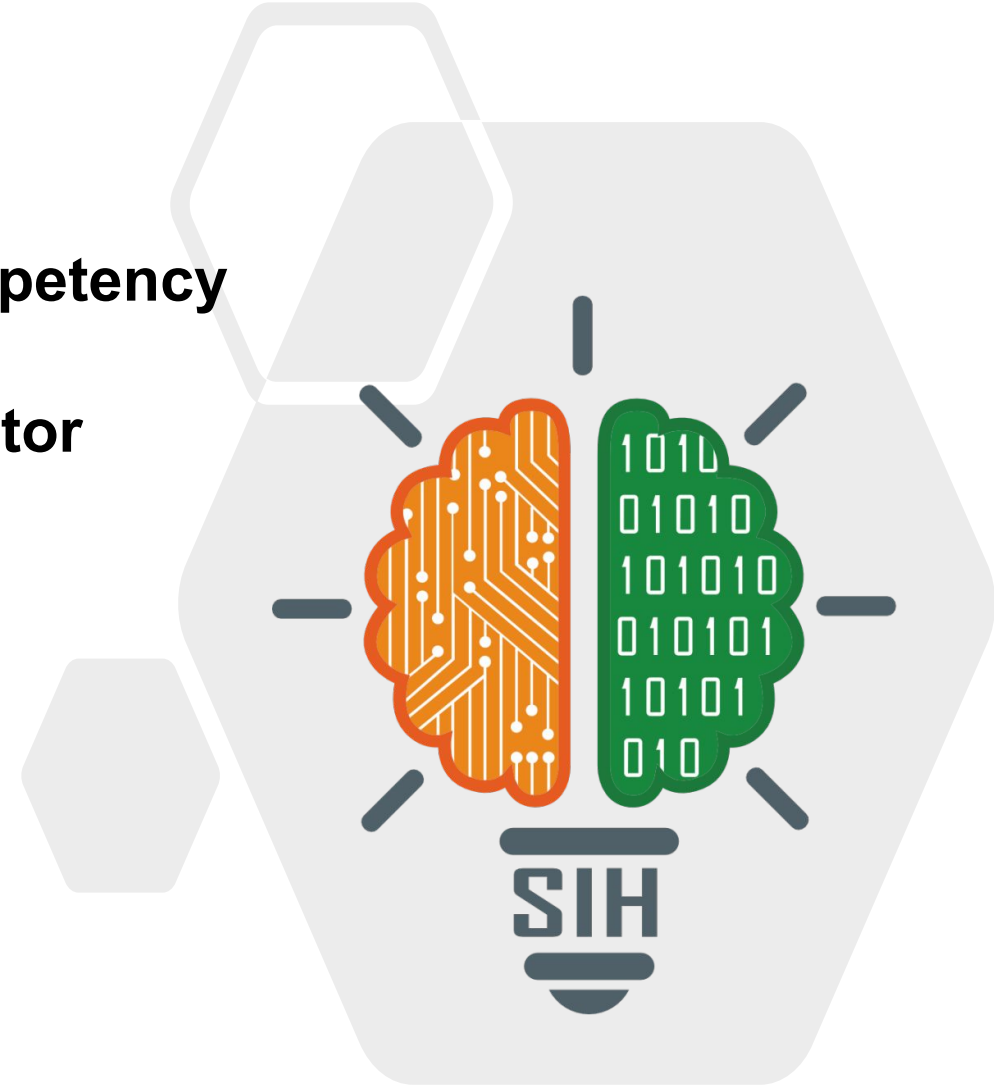


SMART INDIA HACKATHON 2024



TITLE PAGE

- **Problem Statement ID – SIH1628**
- **Problem Statement Title - Smart Competency Diagnostic and Candidate Profile Score Calculator**
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID- 42490**
- **Team Name - PlaceX**



For Job Seekers



Profile-fit Jobs

- Check **probability of getting shortlisted** for a job by matching candidate's profile and skills
- Show **strengths & weaknesses**; Rating on different parameters and **suggest relevant courses** to improve those skills



Customized Job alerts

- Receiving **job alerts** which matches the profile on **email** and **WhatsApp** directly
- Apply these jobs with a single click



Resume Builder

- Allows to select a template, **step-by-step guidance** on filling of each section
- Offers content suggestions like **action verbs, keywords** etc., to customize to a job-role



Interview Preparation

- The tool leverages AI to **analyze responses**, helping users improve their communication and performance skills.
- It provides **personalized feedback** and guidance to users based on **real-time interview simulations**.



Mentorship Service

- Get **guidance from experts** who have cracked a similar job role; can take sessions from them
- Join **networking events** or **engage in discussions** with other professional leaders to share practices / Strategies.

For Employers



Hosting Assessments

- Host **assessment tests** like aptitude, coding and development etc.
- **Personalized Tests** : Based on tuning models on the resume and job description



AI Screening Interviews

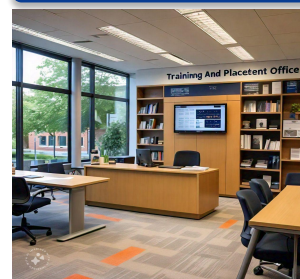
- Powered by cutting-edge AI hiring technology, software analyzes crucial aspects of potential candidates
- **Emotion & Sentiment Analysis** using NLP: Sentiment & communication for **cultural fit**



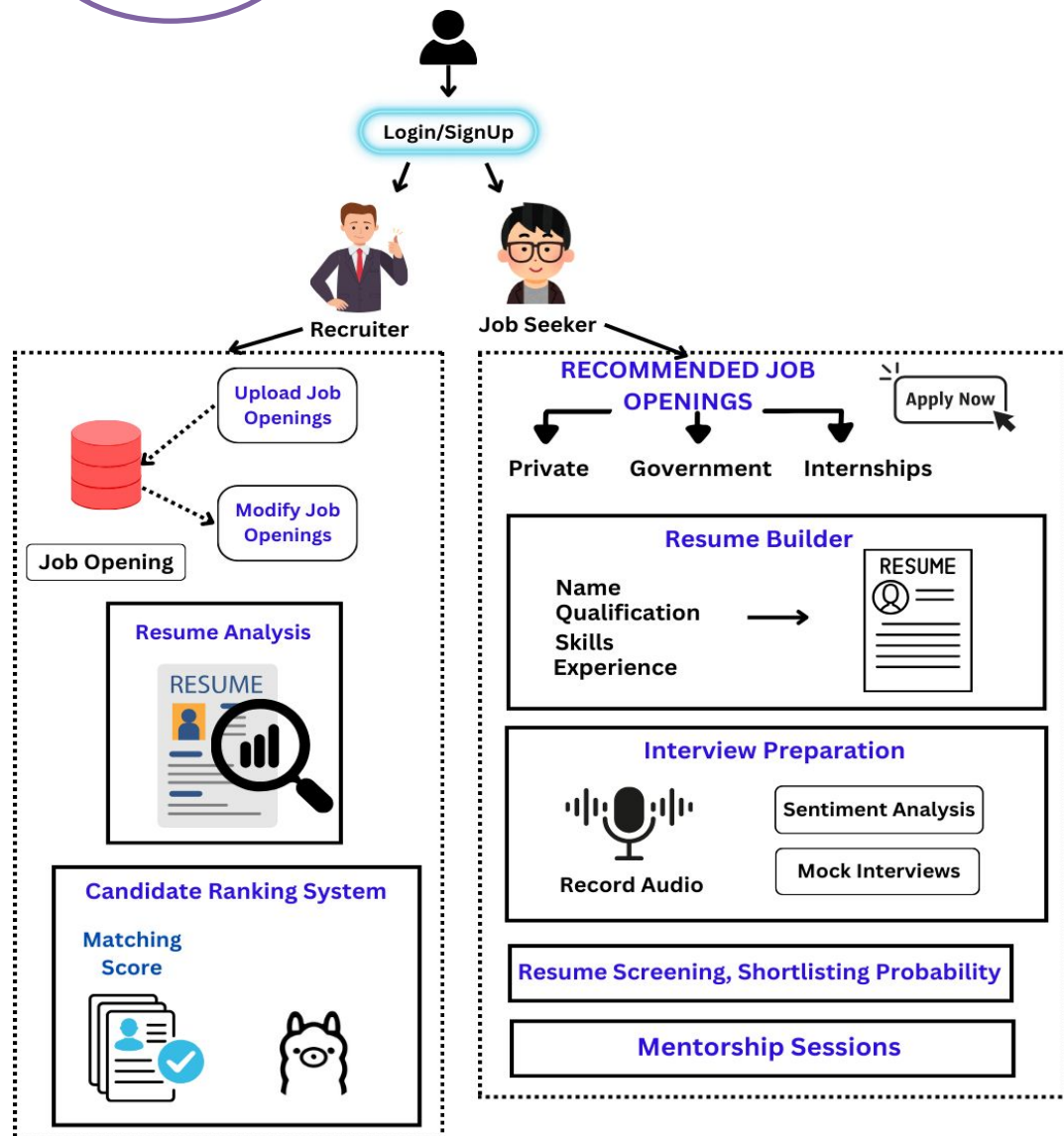
Hotlists

- Get a hotlist with **ranking of candidates** who applied for the role
- **Rating on each parameters** relevant to Job description

Training & Placement Collaboration



- Collaborate with **T&P offices of colleges** initially, to offer services to students
- Can **customize the offerings** like resume builder according to their requirements to bring in more jobs
- **Mock interviews** from experts of community



01 Frontend + Backend
NEXT.JS,
Typescript

02 Styling
TailWind CSS,
Shad CN

03 ORM
Prisma

04 Database
MongoDB

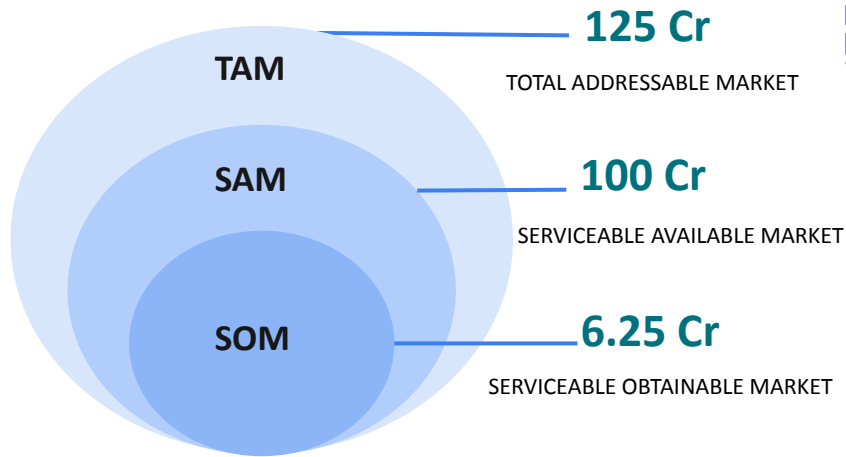
05 Vector Database
Chroma

06 ML Models
llama-3.1,
mxbai-embed-large

07 Voice Model
Hume

08 Other Frameworks
Used
langchain

Market Size

[Calculations](#)

Cost Structure & Revenue Streams

- **Technology and Development Costs:** Development and maintenance of the platform, including UI/UX design, Costs for developing and maintaining AI-based tools for screening interviews.
- **Marketing and Sales Costs:** Marketing campaigns to attract both job seekers and employers, Partnerships with universities, companies etc.,
- **Subscription Fees from Employers:** Subscription fees for access to the platform's features, such as job postings, candidate search, and detailed analytics on candidate assessments.
- Offering **premium services** for job seekers such as advanced resume building, one-on-one career coaching, or interview preparation sessions on a **pay-per-session basis**

Challenges

Bias in AI algorithm

Scalability of AI Models

Differentiating in a Competitive Market

Solutions

Regularly **monitor, audit, and update the AI models** to detect and correct any biases. Use **diverse training data** and implement fairness checks to ensure balanced results. (By using **LoRA and QLoRA techniques**)

Use **cloud infrastructure** to scale the AI capabilities dynamically. Implement **hybrid methods** i.e. not depending totally on LLMs but making use of traditional features such as ATS (Initial shortlisting by ATS and Hotlisting of the top 20% by LLMs)

Emphasize unique features like **shortlisting probability predictions, custom assessments, cultural fit analysis** and community engagement. Highlight **AI-driven insights that go beyond traditional job platforms**.

IMPACT AND BENEFITS

- **Bias Mitigation in AI Tools**

Our commitment to **reduce bias in AI-driven recruitment** processes helps **promote fairness** in hiring. By using diverse training data and actively monitoring for bias, the platform ensures that candidates from all backgrounds have equal opportunities to succeed

- **Inclusive Hiring Practices**

The platform supports recruiters in creating inclusive job postings and evaluating candidates more holistically, reducing the chances of unconscious bias. This fosters a more **diverse workforce**, which is essential for **social equity**.

- **Best Fit Prediction**

AI-driven resume builder and job matching tools help job seekers present themselves effectively to employers, increasing their chances of securing employment. This contributes to **greater workforce participation and economic inclusion**, particularly for individuals facing barriers to employment.

- **Boosting Economy**

We put emphasis on serving freelancers, career changers, and gig workers addresses the growing segment of the workforce that operates outside of traditional employment. By offering tools that cater to these individuals, we help in **reducing the employment disparities** in the gig economy.

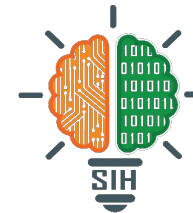
- **Non Traditional Candidate**

By focusing on personalized job matching that considers more than just keywords, we **help underrepresented groups** find opportunities that align with their skills and aspirations, promoting **greater economic participation** across diverse communities.

- **Ethical AI and Employer Branding**

We encourage companies to adopt ethical hiring practices and showcase their **commitment to diversity, sustainability, and employee well-being**. By helping employers build stronger, more inclusive brands, the platform contributes to a corporate culture that values **social responsibility**.

Not my job!



Industry Outlook

19%

Of employers, preferred to hire **Freshers** in 2024

71%

Employability rating for **MBA** graduates which is highest in 2024

65%

Employability rating for **Engineering** graduates which is second highest in 2024

5%

Increase in the **demand** over the past 6 months

Primary Research

Research Sample:

In-depth Interviews:

Job-seekers (22) + Employers (7)

Google Form Responses: 153

69%

Of respondents are **searching jobs** through **online platforms**

68%

Of respondents feel **industry-level skills** are **not taught** in colleges

89%

Of respondents are expecting **profile-relevant jobs** from online platforms

References

Secondary Research

- [Crunchbase](#)
- [SEMrush](#)
- [SimilarWeb](#)
- [Ahrefs](#)
- [CB Insights](#)
- [Owler](#)
- [PitchBook](#)
- [LinkedIn Sales Navigator](#)
- [Elsevier](#)

Primary Research

[Google Form Responses](#)

Student Insights

No information when new job roles are posted by a company

Unable to find desired role easily

Lack of expertise in tools required in JD

Guidance from experts - preparation tips

Employer Insights

Limited vendors to outsource for customized assessments

Not enough applicants when job opening is posted on website

Role-fit candidates do not apply when job is posted

Interviewing large pool takes up a lot of time

Market Voice

"I'm not sure of which all kinds jobs I should apply to"

-Arjun, 4th year Engineering

"I don't hear back from the company after my interview"

- Priya, 2nd year MBA

"Difficult to find the right job role fit candidate with relevant skill-set"

- Rahul, IT Sr.Manager

"Screening higher no.of applicants for limited vacancies is a challenge"

- Ishita, HR Manager